Project 2: Multiverse You

Fork a real decision into many simulated futures and watch them play out on a game-style timeline map

1. Concept

The user describes a real decision ("take the job in another city or stay"). The system forks it into N parallel timelines (start with 12, scale to 50). In each timeline, agents play **you** and the people around you, living 5 years month by month, with random life events. At the end a narrator agent compares all timelines: where regret appeared, what the common failure points were, and which small choices mattered most.

Important framing: this is a reflection tool and a simulation, not a prediction. The UI should say so.

2. How it looks (game style)

- **Multiverse map**: a glowing tree. The trunk is "today", and each branch is a timeline that grows to the right as months pass.
- **Branch colors** show current life state: green thriving, yellow okay, red struggling, grey ended early.
- **Stat bars per timeline**: money, happiness, health, relationships, career.
- **Event pops**: little icons flash on branches (promotion, breakup, illness, windfall, burnout).
- **Time scrubber**: drag through months 0 to 60 and watch every branch change.
- **Click a branch** to read its diary as a month-by-month story.
- **Finale screen**: a ranked list of futures, a regret heat map, and "the 3 moments that mattered most".

3. Using Gemini, NVIDIA and Groq together

Provider	Role here	Base URL	Example model (verify current names)
Groq	Monthly life simulation, the high-volume calls	https://api.groq.com/openai/v1	llama-3.1-8b-instant, llama-3.3-70b-versatile
NVIDIA NIM	Character creation and event generation (creative)	https://integrate.api.nvidia.com/v1	meta/llama-3.1-70b-instruct
Gemini	Narrator, which reads all timelines at the end (long context is its strength)	Google GenAI SDK	gemini-2.5-flash

Use one router module with automatic fallback on 429 or errors. Keys stay in a server-side `.env` file only.

4. Architecture

```
Browser (Canvas/Phaser multiverse map)  <--WebSocket-->  Backend (FastAPI)
                                                           |-- Decision intake agent
                                                           |-- Timeline simulator (N
parallel)
                                                           |-- Event generator
                                                           |-- Narrator / analyst
                                                           |-- LLM router -> Groq /
NVIDIA / Gemini
                                                           |-- SQLite (timelines,
months, events)
```

5. The agent roles

1. **Intake agent**: asks 5 to 8 questions (age, finances, relationships, fears, goals) and builds a structured *You profile* plus the decision options (A, B, or a hybrid C).
2. **World builder**: creates 3 to 5 supporting characters (partner, parent, boss, best friend) with personalities.
3. **Life agent (per timeline)**: each month it receives state, last events and the chosen path, and returns a JSON update.
4. **Event generator**: injects random events with weighted probabilities (job loss 2%, health scare 1%, windfall 1%, new relationship 3%, and so on).
5. **Narrator**: reads all diaries and produces the comparison report.

6. Timeline state (JSON)

```
{
  "month": 14,
  "money": 4200, "happiness": 62, "health": 80,
  "relationships": 55, "career": 70,
  "location": "Pune",
  "events": ["promotion"],
  "diary": "Month 14: ..."
}
```

The life agent returns only deltas and a one-line diary entry, which keeps tokens low and the numbers consistent.

7. Simulation design

- **Step size**: simulate in 3-month chunks first (20 steps for 5 years). Offer "deep mode" at 1 month.
- **Variation between timelines**: each timeline gets a different random seed for events plus a small "behavior twist" (bold, cautious, social, workaholic) so futures really diverge.
- **Anchor path**: roughly half the timelines follow option A, half option B, with a few hybrids.
- **Call count**: 12 timelines x 20 steps = 240 small calls. Run 5 in parallel on Groq, and cache nothing, since variety is the goal.

8. The final analysis

The narrator receives compressed diaries and stats and outputs:

- Ranking of timelines by a weighted life score (the weights come from the user's stated values)
- **Regret map**: months where happiness dropped sharply after a decision
- Common failure patterns ("ran out of savings before month 10")
- The 3 small choices that changed outcomes the most
- A plain-language recommendation, with a clear reminder that it is a simulation

9. Build plan

Day	Goal
1	LLM router with fallback. Test all three keys.
2	Intake agent and the profile JSON.
3	Single-timeline simulator, headless. Verify JSON stability.
4	Parallel timelines, events, behavior twists.
5	Multiverse map visuals with WebSocket streaming.
6	Narrator report, regret heat map, finale screen.
7	Time scrubber, polish, demo recording.

10. Demo script

1. Type a decision live from someone in the audience.
2. Answer the quick intake questions.
3. Press Fork. The tree splits into 12 glowing branches.
4. Branches turn green and red as months tick by, and events pop up.
5. Finale: the narrator says which choices mattered, and the audience argues about it.

11. Safety and honesty

- Show a clear "simulation, not prediction" notice.
- Do not store personal data beyond the session unless the user opts in.
- For serious topics (health, self-harm, major legal or financial decisions), show a gentle suggestion to talk to a real person.
- Avoid fake precision: show ranges and trends, not "you have a 73% chance".

12. Risks and fixes

- **All timelines look the same**: increase event randomness and behavior twists.
- **LLM invents unrealistic numbers**: clamp deltas in code (for example money change limited to a plausible range).
- **Too slow**: simulate in 3-month steps, run in parallel, stream results as they arrive.
- **Tone gets cheesy**: give the life agent a style guide (plain, grounded, no melodrama).

13. Stretch ideas

Voice narrator, shareable "my multiverse" image, a friend mode where two people fork the same decision, and a "what if I had decided this 5 years ago" replay.